Run Android Apps on Your Windows PC

By Daniel A. Begun

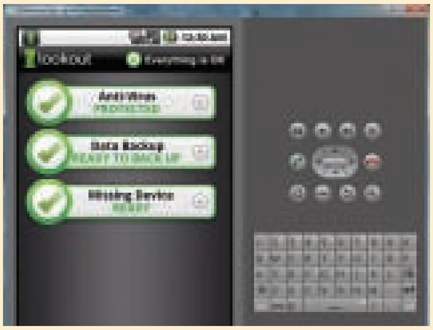
If you're not currently an Android phone user, you may be wondering what all the hoopla is about. If you are truly intrigued by what the Android OS is all about, you can spend some quality time with it, at your leisure, in the comfort of your own home—and for free! All you need to do is download and install some free software, configure a few settings, and you'll have a "virtual" Android phone running on your computer.

First, a few caveats. First and perhaps most important, is that you won't be able to use your virtual Android phone to actually make phone calls. This project is about getting a feel for the Android OS and its user interface, and the opportunity to explore some Android applications—without having to buy anything. Another limitation is that any app that seeks to utilize a phone's camera or GPS will find that these components don't actually exist in the virtual device. Also, any app that uses location services (which provide relevant information based on your current physical location) won't be able to pinpoint where you are. (It is possible to get a virtual Android device to work with an attached camera or GPS, but that sort of deep-developer-level tweaking is well out of the realm of what we're doing here.) You're also going to find that your virtual Android device runs slow—probably much slower than a real Android phone would.

But there are still plenty of things a virtual Android device can do—especially when you start installing apps.

A Few Words About Android Apps

You'll want to keep in mind a few things about Android apps. Unlike the heavily policed iPhone App Store, the Android OS version, the Android Market, has much looser guidelines. In fact, some consider the Android Market to be the wild west of apps: Almost anything goes, and malicious apps are often pulled from the Android Market only after users complain. As to how widespread the potential is for



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malware in the Android Market, the mobile phone software security company, SMobile Systems, recently released a report that estimates that "one in every five applications request permissions to access private or sensitive information that an attacker could use for malicious purposes." It's unlikely that every single one of these apps is requesting this information intentionally for the purpose of sending it back to cybercrooks. In all probability, the majority of these apps are either using this information for legitimate purposes (such as Internet banking) or are just the by-product of developers' sloppy coding. But at least some of these apps may be trying to steal your information and put it in the hands of individuals you probably would rather not have it.

And the Android Market isn't even the only source for downloading and installing Android apps. A number of independent Android apps stores have already been doing business for a while now (such as Softonic, Handango, and GetJar), and new ones are starting to pop up (such as AndSpot, SlideMe, and AndAppStore) now that the popularity of Android phones is increasing. How these independent sources will mind the cybergates still remains to be seen.

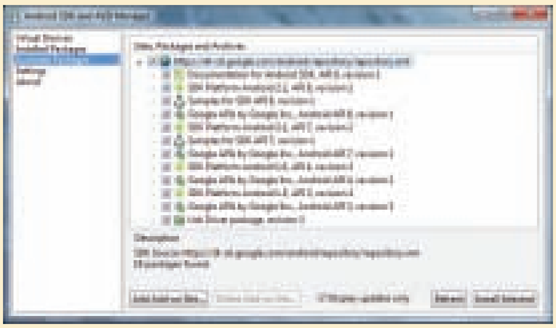
The bottom line is that you need to be careful what you install on an Android device—even a virtual one—especially if you've already entered some personal information, such as your Google account credentials. How do you know if an app is really malware in disguise? Without having some sort of security software installed, it might be hard to tell. We installed the no-cost Lookout Mobile Security FREE (beta) app on several of my virtual Android devices (and my real Android phone), and as far as we can tell, it does what it is supposed to: protect against malware.

In the following pages we show you how to create and use a virtual Android device on a Windows PC, as well as how to download and install Android apps from the independent Android apps stores. This is all done with the latest iteration of the Android OS: 2.2, which is also known as Froyo (Froyo is shorthand for "frozen yogurt"—all of the Android OS codenames are named after desserts). And we finish up this primer with what you need to know to set up an Android 1.6 virtual device with a working Android Market.

Installing the Android SDK

All of the examples we provide here were done on a system running Windows 7; but they should be just as applicable for virtually any system running 32-bit Windows XP or 32- or 64-bit Windows Vista. There are even versions of the Android SDK that run on Intel-based Macs loaded with Mac OS X 10.5.8 and later and some versions of Linux, but the steps to getting this to work on these other operating systems differ somewhat from you'll find in this guide.

The first thing you need to do is to make sure that Java is installed on your



The download components of the Android SDK